Autonomous Floor Cleaners

A cost and feasibility review for Greeneville City Schools custodial operations

Prepared for Jason B. Horne, Assistant Director of Schools for Administration | August 22, 2026 | Research review, not a purchase recommendation

1. Executive Summary

Autonomous floor scrubbing is further along in K-12 than autonomous mowing, and the evidence base is meaningfully better. Multiple named districts have deployed at scale, at least one deployment is documented in a public board record with dollar figures, and one is covered by independent government-technology journalism with usage data. This is a category where GCS can look at what peers actually did rather than at what vendors claim.

The core finding is that the value here is almost entirely about labor capacity, not labor cost. Every district found in this review kept its custodial headcount and redeployed the recovered hours. The savings show up as reduced overtime, fewer missed routes, less reliance on substitute and temporary labor, and detail work that finally gets done, not as a smaller payroll. If the GCS business case is built on eliminating custodial positions, the peer evidence does not support it.

What the money looks like

Item	Range found	Source type
Compact autonomous scrubber, K-12 sized	$27,500 to $35,833	Reseller published pricing
Mid-size autonomous scrubber	$41,820 to $51,049	Reseller and distributor pricing
Rider-class autonomous scrubber	$77,807	Distributor pricing
Industrial rider class	$60,000 to $85,000+	Category survey
Robots as a service, monthly	$450 to $1,200 per robot	Category survey
Robots as a service, low end all-inclusive	about $15 per day	Vendor press release
Annual consumables per robot	$500 to $1,000	Category survey
Annual service plan	10% to 15% of purchase price	Category survey
Battery replacement, every 3 to 5 years	$2,000 to $5,000	Category survey
Robot total operating expense per year	$12,000 to $18,000	Trade press
US K-12 custodian, fully burdened	$55,000 to $70,000 per year	Trade press

Full citations are in Section 12. Source type is stated because reliability varies. See Section 2.

Bottom line up front

- Floor care is reported to consume 30 to 40 percent of custodial labor, and 50 to 75 percent of that floor work is automatable. That is the size of the prize, and it is large.

- A single machine covers roughly 15,000 to 25,000 square feet of repeatable hard floor per day, which in practice means one machine handles one school's corridors, cafeteria, and gym approach, not a whole school.
- Realistic recovery is around four hours of custodial time per day per building, which is the figure both Council Bluffs and Greenville County independently reported.
- GCS cannot compute payback yet. Cleanable hard floor square footage is not recorded anywhere in district files, and it is the primary driver.
- One caution specific to GCS: three gym floors were resurfaced recently. Autoscrubbers should not be used on newly resurfaced wood until the finish has been built up over one to three years of recoating. Those gyms must be excluded from any pilot route.

2. How to Read the Evidence in This Market

The evidence quality here is better than in autonomous mowing, but the same grading discipline applies. Several widely quoted figures, including the claim that a human cleaner costs four to seven times what a robot costs, come from reseller marketing and compare a full-time employee against a machine that cleans only hard floors for a few hours a night. That is not a like-for-like comparison and it should not be repeated.

Tier	What it is	Examples in this review
Tier 1	Public board records, government reporting, independent journalism	Council Bluffs board action, Spokane via GovTech
Tier 2	Manufacturer specifications, distributor list prices, professional standards bodies	Tennant and Karcher specs, HD Supply pricing, APPA standards
Tier 3	Reseller blogs, category surveys, vendor case studies	Per-machine ROI examples, RaaS ranges, savings multiples

3. What the Technology Actually Is in 2026

Three machine types get grouped under autonomous cleaning, and only one of them is relevant to a district's biggest labor line.

Autonomous scrubbers, the main event

A scrubber lays down water and detergent, agitates with brushes or pads, and vacuums the solution back up through a squeegee. An autonomous scrubber does that on a route it has been taught, without a person walking behind it. This is the machine that matters, because wet floor care in corridors, cafeterias, and commons is the most time-consuming and least skilled part of the nightly custodial routine.

The dominant platform is Brain Corporation's BrainOS, which powers the Tennant AMR line and several others. Competing platforms include Avidbots, Karcher's KIRA, Nilfisk Liberty, and a growing set of lower-cost entrants including Gausium and CenoBots.

Autonomous vacuums and sweepers

Smaller, cheaper machines for carpet and dry debris. Useful in carpeted corridors and libraries, but they address a smaller share of custodial time and generally have weaker ROI in K-12 than scrubbers do.

What none of them do

No current-generation machine operates an elevator independently, cleans a restroom, cleans a classroom well, moves furniture, or handles stairs. Every one of those remains manual. This is the single most important thing to understand before modeling savings, because it caps the achievable labor reduction far below what the marketing implies.

4. Vendor Landscape and Published Pricing

Vendor and model	Class	Published price	Key specification	Model
Sproutmation L3 + WS3	Compact	$27,500	Tighter corridors	Purchase
Sproutmation L4	Compact	$35,833	Cafeteria, gym, corridor loop	Purchase
CenoBots L50	Mid	$41,820 + $4,667 station	Fieldhouse scale	Purchase
Gausium Scrubber 50 Pro	Mid	$43,000 or $1,795/mo	RaaS available	Purchase or RaaS
Tennant X4 ROVR	Mid walk-behind	$51,048.67	10 gal, 20,000 sq ft per tank, 2.5 hr	Purchase
Tennant T7AMR	Mid rider	Quote	26 in path, 29 gal, 45,760 sq ft	Purchase
Tennant T380AMR	Rider	$77,806.67	20 in disk, 20 gal, up to 4.5 hr	Purchase
Karcher KIRA B 50	Mid walk-behind	Quote	22 in path, ~24,750 sq ft/hr, 3.5 hr	Purchase
Avidbots Neo 2	Industrial	$60,000 to $85,000+	Up to ~42,000 sq ft/hr	Purchase
Gausium Scrubber 75P	Industrial	$95,880 or $2,013/mo	RaaS available	Purchase or RaaS
ICE Cobotics Cobi 18	Compact	$16,000 over 36 months	All-inclusive at about $15/day	RaaS only
Nilfisk Liberty SC50 / SC60	Mid	Quote	98% to 99.5% coverage claimed	Purchase

Prices are as published by distributors and resellers in 2025 and 2026. Every one is a starting point for a quote. Tennant pricing shown is HD Supply distributor list, which is typically above a negotiated public-sector or cooperative price.

The recurring costs that decide the answer

On a purchase, the machine price is roughly half the five-year cost. The rest is service, consumables, and eventually a battery.

- Annual service plan, typically 10 to 15 percent of purchase price. On a $40,000 machine that is about $4,000 per year.
- Consumables, $500 to $1,000 per robot per year for squeegee blades, brushes, and filters.
- Battery replacement, $2,000 to $5,000, on a 3 to 5 year cycle for standard lithium-ion.
- Mapping and deployment, which is a real labor cost at first setup and again whenever a building is reconfigured.
- Water, detergent, and the labor to fill and dump tanks, which does not disappear and is a recurring complaint in the Spokane account.

The all-inclusive robots-as-a-service alternative bundles all of that into one number. ICE Cobotics prices the Cobi 18 at $16,000 across 36 months, about $15 per day, including fleet management, analytics, support, repairs, replacement parts, and consumables. The broader category runs $450 to $1,200 per robot per month depending on machine size and service level.

5. What Districts Have Actually Done

Spokane Public Schools, Washington

The largest documented K-12 deployment and the best-sourced one. Spokane purchased 14 Tennant autonomous floor cleaners for $1.08 million, placing one at each middle and high school. Warranty was two years on labor and three years on parts. The district set a minimum usage requirement of three times per week, which is a detail worth copying because underuse is the most common way these machines fail to deliver.

Metric	Reported
Units deployed	14
Total cost	$1.08 million
Cost per unit, implied	About $77,000
Hours logged at time of reporting	Over 1,500
Square footage cleaned	Over 16.5 million
Positions eliminated	None
Minimum required usage	3 times per week

The superintendent framed the purchase as covering staffing shortages, specifically staff calling out sick with insufficient substitutes available, rather than as labor replacement. The teachers association president supported it on the grounds that the robots still require staff to monitor them and represent workload reduction rather than job replacement. A custodial assistant quoted in the coverage was more measured, saying the robot had not yet made his job faster and comparing it to caring for a baby. The reported challenges were regular monitoring, water refills, and manual intervention when the machine gets stuck.

The implied $77,000 per unit is high relative to published distributor pricing for most models, which suggests the total included deployment, training, service, or a higher-end rider configuration. Do not use $77,000 as a planning figure without knowing what was in it.

Council Bluffs Community School District, Iowa

The most useful example for GCS because it is a public board record of a cautious first step rather than a fleet purchase. On March 12, 2025, the board approved a four-year lease-to-own agreement for two units, a Cinnabot L50 and a Spot Cleaner 50, totaling approximately $90,000. The lease included mechanical support for the full term.

The reported operational result was that the machines freed roughly four hours of custodial time per day for one custodian in the demonstration building. Board members asked about cost and equipment lifespan, battery life was estimated at ten years, no objections were raised, and the motion passed by roll call. One board member noted the district would run a full year of data before broader rollout.

That last point is the model. One building, one year, real data, then a decision.

Greenville County Schools, South Carolina

A large district, 900 custodial employees across 85 campuses, running a robotic sweeper and a robotic scrubber. Projected time savings were reported at four to six hours per day. The independent arrival at roughly the same four-hour figure as Council Bluffs, in a district of a completely different size, is the closest thing this category has to a validated planning number.

De Pere High School, Wisconsin

A single-building deployment of an AI-powered autonomous floor scrubber at a reported equipment cost of $89,000. No outcome data published.

The common thread

All four districts run what the trade press calls a cobot model, where the machine handles repetitive open-floor work and a person handles everything else. Leadership in every case framed automation as extending staff capability, with custodians redirected toward high-touch surfaces, restrooms, and exterior work while robots take corridors, cafeterias, and gyms. Not one district found in this review eliminated custodial positions.

6. The Labor Question

This decision has to be settled before equipment is priced, because it determines whether the project produces cash or capacity.

	Reduce headcount	**Redeploy hours**
Where savings appear	Payroll, immediately	Overtime, subs, contract cleaning, deferred detail work
Peer precedent	None found in K-12	All four districts reviewed
Staff and association reaction	Adversarial	Supportive where framed as workload relief
Realistic given vacancies	Only through attrition	Yes
What it actually buys	A smaller payroll line	Cleaner buildings and fewer missed routes

Sizing the opportunity honestly

The most quoted structural figure is that routine floor care accounts for 30 to 40 percent of custodial labor, and that automating 50 to 75 percent of floor work is achievable. Multiplying those out gives a theoretical ceiling of roughly 15 to 30 percent of total custodial labor. That ceiling is theoretical because it assumes every building has enough contiguous hard floor to justify a machine, which is not true of a small elementary school.

The observed figure across two independent districts is about four hours per day per building. At a fully burdened custodial cost in the $55,000 to $70,000 range, four hours a day across a 180 to 200 day school year is meaningful but is not a full position. It is roughly half of one.

Do not multiply four hours per day by every GCS building. That figure came from demonstration buildings, which are chosen because they suit the machine. A small elementary with carpeted corridors will not produce it.

The APPA staffing frame

APPA, the higher education facilities association, publishes the five-level cleaning standard that most districts use as a reference. The relevant point for this analysis is that APPA staffing is derived from cleanable square footage divided by a productivity standard, with a common divisor around 25,000 square feet per custodian per day, and that the divisor shifts substantially by space type and target cleanliness level. A custodian can cover roughly double the area at Level 3 that they can at Level 1.

This gives GCS a second, better way to frame the decision. Rather than asking what the robot saves, ask what APPA level the district is currently staffed to and what level it wants to be at. If GCS is staffed below its target level, automation closes the gap without new positions. That is a stronger and more honest argument than a payroll savings claim, and it is one a board can act on.

7. What Drives the Payback Math

The accompanying workbook, Autonomous Floor Cleaner ROI Model.xlsx, implements this model with live formulas. In order of leverage:

1. Robot-eligible hard floor square footage per building. Not gross square footage, not cleanable square footage. Only the contiguous hard floor the machine can actually run without a person moving furniture. This is usually a fraction of the building.
2. Nightly minutes currently spent on that specific floor work. This is what converts square footage into dollars, and it has to be measured, not inferred.
3. Nights of operation per year. School buildings run roughly 180 to 200 instructional days plus summer work, which is materially fewer than the 300-plus nights a retail or industrial site gets. This is the biggest single reason K-12 payback is slower than commercial payback, and it is rarely mentioned in vendor material.
4. Fully burdened custodial hourly rate. Use the GCS classified rate, and coordinate with the active 2026-2027 Classified Salary Study rather than pulling a rate independently.
5. Carpet share. Every square foot of carpet is a square foot the scrubber does not touch and cannot be counted in the ROI.

6. Number of floors and elevator dependency. A machine that cannot move itself between floors needs a person to transport it, which eats directly into the hours saved.
7. Current overtime and substitute or temporary cleaning spend. In a redeployment model this is where the savings actually land, and if the district cannot name a current dollar figure here, it will not be able to prove savings later.

A worked illustration, not a GCS projection

A published reseller example for a mid-size high school runs as follows: $15,593 of annual labor value protected, $7,167 per year of machine cost amortized across five years, $1,800 per year of service and consumables, producing net annual savings of $6,568 and a payback of 21.8 months. That implies a machine cost near $35,800 and roughly 18,000 square feet cleaned per day.

Tier 3 figure, published by a reseller. It also assumes a 260-night operating year implicitly. Re-run it at 190 nights, which is closer to a school reality, and the labor value protected drops by about a quarter and payback stretches past two and a half years. The workbook lets you set the night count directly for this reason.

8. What Breaks It

Carpet

Autonomous scrubbers do hard floors. Carpet extraction and vacuuming stay manual and are not part of the ROI. Older school buildings frequently have carpeted corridors, libraries, and offices, and each one shrinks the addressable area. Survey carpet share per building before sizing anything.

Wood gym floors, and a specific GCS caution

This is the most important operational warning in this report. Automatic scrubbers should not be used on newly installed or newly sanded wood floors. It generally takes one to three years of recoating to build up enough finish to protect the wood well enough for autoscrubbing. Additionally, machine weight must not exceed the flooring weight limit. Rider scrubbers can exceed those limits and cause indentations, and suspended floor systems can crack.

GCS resurfaced three gym floors in the recent maintenance cycle. Those specific floors must be excluded from any autonomous scrubbing route until the finish has been rebuilt, and the flooring contractor or manufacturer should be asked in writing what the weight limit is before any machine touches them. Getting this wrong damages an asset that costs far more than the robot.

Restrooms and classrooms

Both are poor automation fits. Restrooms have complex geometry, fixtures, and manual detail requirements. Classrooms have low square footage relative to furniture density. These are the spaces custodians should be redeployed into, not the spaces the robot takes.

Elevators and multi-floor buildings

Current machines cannot operate an elevator independently. Multi-floor deployment requires a supervised transport protocol, meaning a person moves the machine floor to floor. That reduces but

does not eliminate corridor cleaning labor, and it should be modeled explicitly for Greeneville High and Greeneville Middle if either has multi-floor corridor routes.

Clutter, which schools generate more of than any other setting

Objects left in corridors interrupt robot runs. In a school that means backpacks, band cases, rolling carts, stacked chairs, art projects, athletic equipment, and cafeteria tables that were not folded. A robot that stops halfway through its route at 9 p.m. and waits for a human produces zero savings that night. Any deployment needs a written clear-corridor expectation communicated to staff, and buy-in from the principal at the pilot building, or the machine will simply not complete routes.

Water, tanks, and babysitting

The recurring practical complaint in the Spokane coverage was monitoring, water refills, and freeing the machine when it gets stuck. Tank capacity sets a hard limit on how much floor gets done per fill. The Tennant X4 ROVR covers about 20,000 square feet on a 10 gallon tank. A larger tank means fewer trips to the sink, and on a walk-behind class machine that is a real productivity variable, not a spec-sheet footnote.

The underuse failure mode

The most common way this investment fails is that the machine gets used twice a month instead of nightly. Spokane addressed this directly by setting a minimum three-times-per-week usage requirement. Any GCS deployment should set and monitor a usage floor from day one, and the fleet management dashboards that come with these platforms make that trivial to track.

9. Procurement Path for GCS

Threshold

Greene County had a 2020 census population of 70,152, above the 40,000 line in state purchasing law. Under TCA 49-2-203, an LEA in a county of 40,000 or more may purchase in the open market without newspaper advertisement below $25,000, based on at least three competitive bids whenever possible. At or above that threshold, competitive sealed bids solicited by newspaper advertisement are required unless the district follows the local governing body's purchasing procedures.

This is a harder constraint here than it is for mowers. Most credible autonomous scrubbers sit above $25,000. Only the compact class and the RaaS subscriptions stay under it.

Confirm the current threshold with the district CFO before it appears in a board document.

Three routes above the threshold

8. Competitive sealed bid. Slowest, but clean, and it produces genuine price competition in a category where distributor list prices are clearly negotiable.
9. Cooperative contract. Riding a Sourcewell, OMNIA Partners, or NASPO ValuePoint contract satisfies competitive purchasing without running a bid. Check each for the specific manufacturer before assuming coverage.

10. Lease or robots-as-a-service. Council Bluffs used a four-year lease-to-own at roughly $90,000 for two units. RaaS at $450 to $1,200 per robot per month keeps annual spend under the threshold and shifts service, parts, consumables, and obsolescence to the vendor. Verify with the district CFO how a multi-year lease or subscription is treated for threshold and for SBITA reporting purposes before committing.

Purchase versus lease versus service

Model	Fits when	Watch for
Purchase	Capital available, service capacity in house	Service plan at 10 to 15 percent per year is permanent
Lease to own	Preserving capital, wanting mechanical support included	Total paid exceeds purchase price, and it may be an SBITA
RaaS subscription	Uncertain fit, wanting everything bundled, staying under threshold	Never converts to an asset, cost continues indefinitely

For a first GCS deployment, the strongest argument is a subscription or short lease on one machine at one building. It stays under the bid threshold, it bundles service and consumables so the true operating cost is visible immediately, and it can be stopped at the end of the term if the fit is wrong. The single biggest risk in this category is buying a machine that turns out not to match the building, and a subscription makes that mistake cheap.

10. Recommended Pilot Design

11. Pick the building with the most contiguous hard floor and the fewest carpeted corridors. Greeneville High or Greeneville Middle is the likely candidate on size, but the carpet survey decides it, not the enrollment.
12. Measure robot-eligible hard floor square footage, not gross square footage. Walk the actual route with a wheel. Exclude classrooms, restrooms, carpet, and anything requiring furniture moves.
13. Exclude every recently resurfaced gym floor from the route, and get the flooring weight limit in writing before considering any gym.
14. Baseline before buying. For four consecutive weeks, log the minutes custodians spend on that exact route, separately from all other work. This is the number the whole model rests on and it costs nothing to collect.
15. Pull current overtime and substitute or temporary cleaning spend for that building. In a redeployment model this is where savings must show up, so the before figure has to exist.
16. Get quotes on one compact-class machine, structured as a subscription or short lease, sized to the measured square footage. Check Sourcewell and OMNIA for the specific manufacturer.
17. Set a minimum usage floor before the machine arrives. Three runs per week, following Spokane, is a reasonable starting standard, and track it on the vendor dashboard.
18. Get the building principal and the head custodian into the design conversation before delivery, and issue a written clear-corridor expectation to staff.
19. Run one full school year. Log route completion rate, interventions per run, water fill trips, downtime, and consumable spend.

20. Compare logged custodial hours and overtime against the pre-deployment baseline. Then decide on expansion, following the Council Bluffs pattern of a full year of data before broader rollout.

11. Data GCS Must Collect Before Any Projection

None of the following is in district files. The district operations tracker lists square footage cleaned daily as an open item requested from operations in May 2026 and still outstanding. Until these exist, any dollar projection is an assumption in a spreadsheet.

Data point	Owner	Why it matters
Robot-eligible hard floor sq ft, by building	Operations director	Primary driver of the entire model
Carpet share by building	Operations director	Directly shrinks the addressable area
Number of floors and elevator dependency	Operations director	Determines transport labor added back
Nightly minutes on floor care, by building	Operations director	Converts square footage into dollars
Custodial nights of operation per year	Operations director	The reason K-12 payback lags commercial
Fully burdened custodial hourly rate	HR director / CFO	Coordinate with the Classified Salary Study
Current custodial overtime spend, by building	CFO	Where redeployment savings actually land
Current substitute and temp cleaning spend	CFO	Second place savings land
Current custodial vacancies and time to fill	HR director	Decides whether this is savings or capacity
Gym floor resurfacing dates and weight limits	Operations director	Hard exclusion rule, protects a costly asset
Target APPA cleaning level, and current actual	Operations director	Reframes the ask from savings to service level

12. Verdict

Autonomous floor scrubbing is the stronger of the two opportunities reviewed. The technology is proven in K-12 at scale, the peer evidence is real and traceable, the labor problem it addresses is the one GCS actually has, and two independent districts arrived at the same roughly four hours per day per building result. That convergence is worth more than any vendor projection.

The honest framing is that this is a capacity purchase, not a savings purchase. It buys cleaner buildings, fewer missed routes, less overtime, and custodial time redirected to work that is currently not getting done. It does not buy a smaller payroll, and no district found in this review claimed otherwise. Framed that way it is defensible to a board and to staff. Framed as a labor reduction it will be neither accurate nor well received.

Recommended next step: request the hard floor square footage and carpet survey from the operations director, run the four-week nightly floor-care time study at the largest hard-floor building, pull current custodial overtime by building from the district CFO, and populate the accompanying workbook. Then price a single-machine subscription pilot under the open-market threshold.

13. Sources

Tier 1: Government, board record, and independent reporting

1. GovTech, Spokane Public Schools automates floor cleaning with robots (14 units, $1.08M, 1,500+ hours, 16.5M sq ft, staffing and union response) https://www.govtech.com/education/k-12/spokane-public-schools-automates-floor-cleaning-with-robots

2. Council Bluffs Community School District board action, March 12 2025, four-year lease-to-own approximately $90,000 for two units https://citizenportal.ai/articles/6640118/Board-approves-four-year-lease-for-autonomous-floor-scrubbers-to-reduce-custodial-hours

3. US Census Bureau QuickFacts, Greene County, Tennessee (2020 population 70,152) https://www.census.gov/quickfacts/greenecountytennessee

4. Tennessee Code 49-2-203, LEA purchasing thresholds https://law.justia.com/codes/tennessee/title-49/chapter-2/part-2/section-49-2-203/

5. APPA, cleaning operations and the five levels of clean https://www.appa.org/cleaning-operations

6. Boise State University, APPA custodial standards reference https://www.boisestate.edu/operations/facilities/fom-departments/custodial-services/appa-custodial-standards/

Tier 2: Manufacturer specifications and distributor pricing

1. Tennant T7AMR robotic floor scrubber specifications https://www.tennantco.com/en_us/1/machines/scrubbers/product.t7amr.robotic-floor-scrubber.2000056.html

2. Tennant X4 ROVR autonomous floor scrubber specifications https://www.tennantco.com/en_us/1/machines/scrubbers/product.x4-rovr.autonomous-floor-scrubber.2000084.html

3. HD Supply distributor pricing, Tennant X4 ROVR and T380AMR https://hdsupplysolutions.com/p/tennant-company-x4-rovr-autonomous-floor-scrubber-p412710

4. Karcher KIRA B 50 autonomous scrubber dryer specifications https://www.kaercher.com/int/professional/cleaning-robots/kira-b-50-15330020.html

5. Nilfisk Liberty robotic floor scrubbers https://www.nilfisk.com/en-us/professional/campaigns/intelligent-cleaning/

6. ICE Cobotics Cobi 18 autonomous scrubber and subscription structure https://us.icecobotics.com/equipment/cobi-autonomous-scrubber

7. Brain Corp, Tennant X4 ROVR specification announcement https://www.braincorp.com/resources/tennant-announces-full-specification-capabilities-of-x4-rovr-autonomous-floor-cleaning-machine-its-first-purpose-built-robotic-scrubber

Tier 3: Trade press, reseller, and category surveys

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2. CleanLink, autonomous floor cleaners close school staffing gaps (Greenville County, Spokane, De Pere) https://www.cleanlink.com/news/article/Autonomous-Floor-Cleaners-Close-School-Staffing-Gaps--32735

3. Sproutmation, school cleaning robot buyer's guide for K-12 and universities (model pricing, sq ft per day, ROI worked example) https://sproutmation.com/blog/school-university-cleaning-robot

4. Commercial Robot Vacuums, the real cost of robot cleaners 2026 (RaaS ranges, service plans, consumables, battery replacement) https://commercialrobotvacuums.com/real-cost-of-commercial-robotic-floor-cleaners/

5. GrabaRobot, commercial cleaning robot comparison 2026 (Avidbots, Gausium pricing) https://www.grabarobot.com/blog/commercial-cleaning-robot-comparison-2026/

6. QHF Sports, the truth about gym floor auto scrubber use (finish build-up and weight limits) https://qhfsports.com/maintenance-faqs/the-truth-about-gym-floor-auto-scrubber-use/

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9. De Anza College, custodial and grounds square footage per person https://www.deanza.edu/gov/campus_facilities/documents/January_28_20_Custodial_Grounds_Cleaning.pdf

Internal GCS sources

1. District operations tracker, square footage cleaned outstanding as of May 2026

2. Director of Schools briefing document, maintenance summary noting three gym floors resurfaced